LECTURE 8

Visualization

Visualizing distributions and relationships; applying transformations and KDEs

Presented by Dr. Devan Becker, Data200

Adapted from Data 100, Summer 2023 @ UC Berkeley by Bella Crouch and Dominic Liu

Content credit: Acknowledgements

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Announcements

- Working on fixing A1 Autograder
- Project out soon!
- A2 may be delayed.

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Goals for this Lecture

Understand the theories behind effective visualizations and start to generate plots of our own

- The necessary “pre-thinking” before creating a plot
- Python libraries for visualizing data

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Agenda

- Goals of visualization
- Visualizing distributions
- Kernel density estimation
- Visualizing relationships
- Transformations

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Goals of Visualization

- **Goals of visualization**
- Visualizing distributions
- Kernel density estimation
- Visualizing relationships
- Transformations

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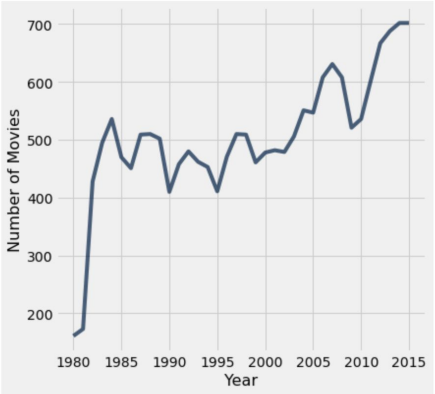
Visualizations in Data 100

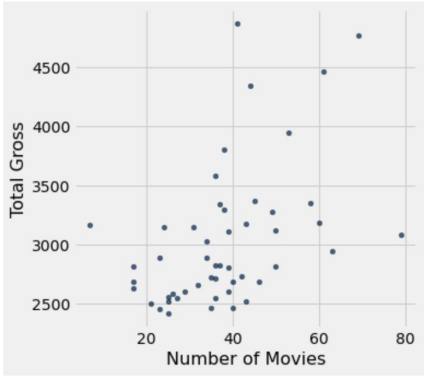
You worked with many types of visualizations throughout Data100.

```
# The call is
# t.plot(x_label, y_label)

movies.plot('Year', 'Number of Movies')
```



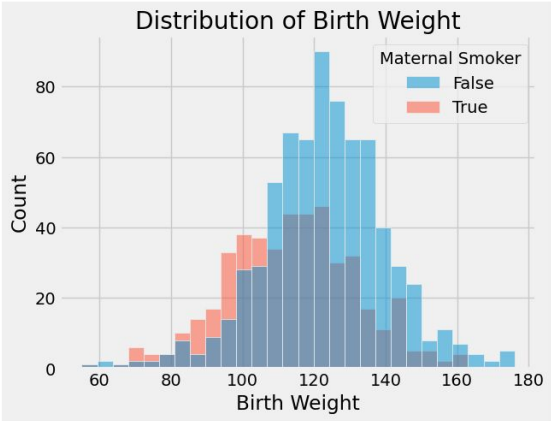
Line plot

```
# The call is
# t.scatter(x_label, y_label)

actors.scatter('Number of Movies', 'Total Gross')
```



Scatter plot

```
import seaborn as sns
sns.histplot(births_df, x="Birth Weight", hue="Maternal Smoker");
plt.title("Distribution of Birth Weight");
```



Histogram

What did these achieve?

- Provide a high-level overview of a complex dataset
- Communicated trends to viewers

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Goals of Data Visualization

Goal 1: To **help your own understanding** of your data/results.

- Key part of exploratory data analysis.
- Summarize trends visually before in-depth analysis
- Lightweight, iterative and flexible.

Goal 2: To **communicate results/conclusions to others**.

- Highly editorial and selective.
- Be thoughtful and careful!
- Fine-tuned to achieve a communications goal.
- Considerations: clarity, accessibility, necessary context

What do these goals imply?

Visualizations aren't a matter of making "pretty" pictures.

We need to do a lot of thinking about what stylistic choices communicate ideas most effectively.

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Goals of Data Visualization

What do these goals imply?

Visualizations aren't a matter of making "pretty" pictures.

We need to do a lot of thinking about what stylistic choices communicate ideas most effectively.

First half of today's lecture: choosing the "right" plot

- Introducing plots for different variable types
- Generating these plots through code

Second half of lecture: stylizing plots appropriately

- Smoothing and transforming visual data
- Providing context through labeling and color

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Visualizing Distributions

- Goals of visualization
- **Visualizing distributions**
- Kernel density estimation
- Visualizing relationships
- Transformations

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Distributions

A distribution describes…

- The set of values that a variable can possibly take
- The frequency with which each value occurs

…for a **single** variable

Example: distribution of students across discussion sections in Data 100

- The list of discussion sections (10-11 am, 11-12 pm, etc.)
- The number of students enrolled in each section

In other words: how is the variable distributed across all out its possible values?

This means that percentages **should sum to 100%** (if using proportions) and counts should **sum to the total number of datapoints** (if using raw counts).

Let's see some examples.

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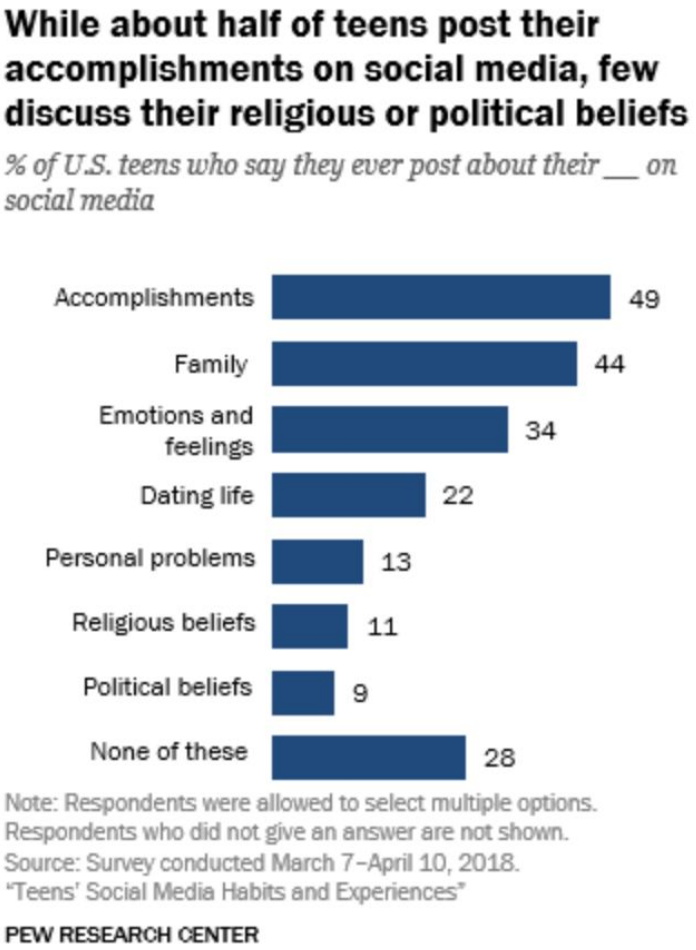

Does this chart show a distribution?

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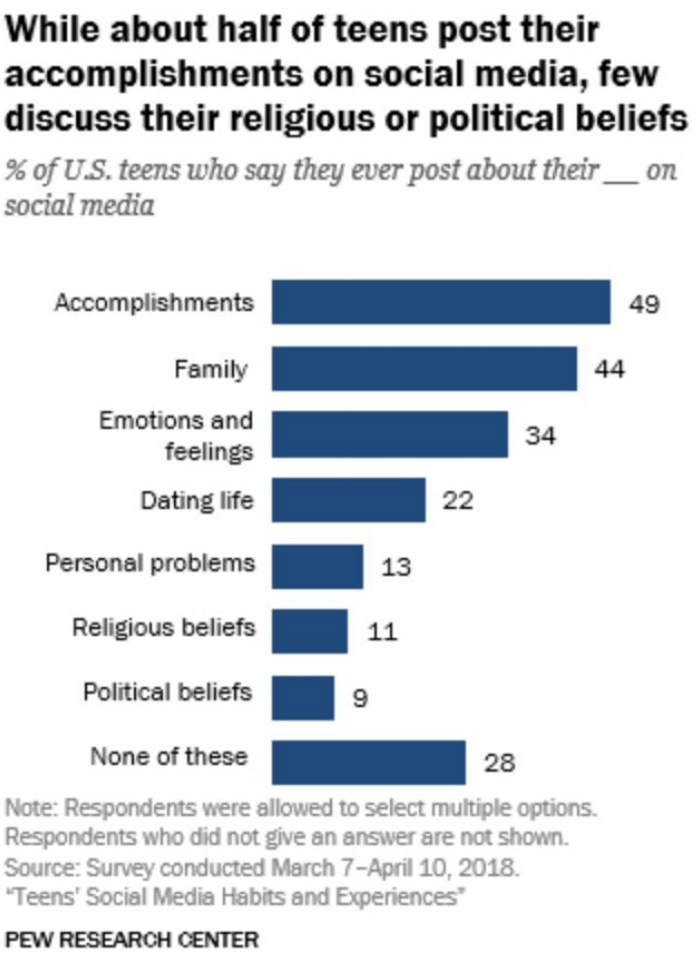

Does this chart show a distribution?

No.

- The chart does show percents of individuals in different categories!
- But, this is not a distribution because individuals can be in more than one category (see the fine print).

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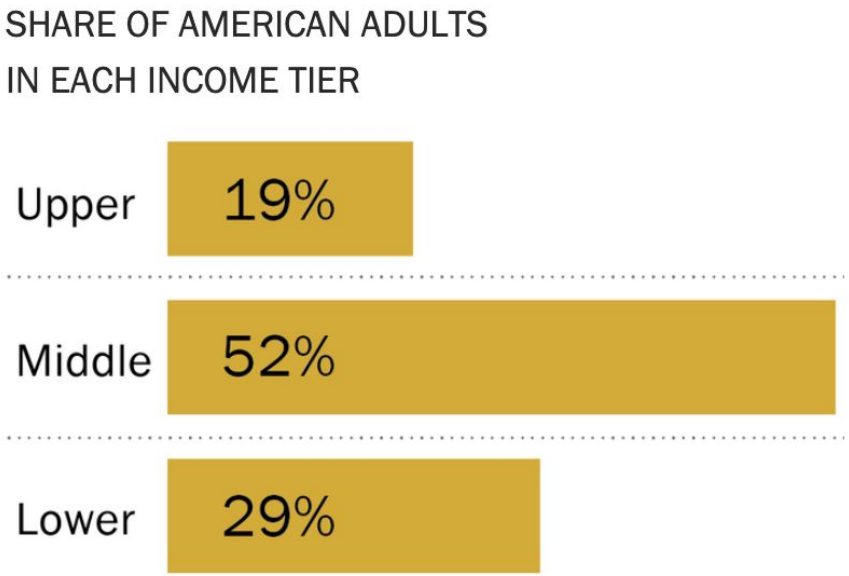

Does this chart show a distribution?

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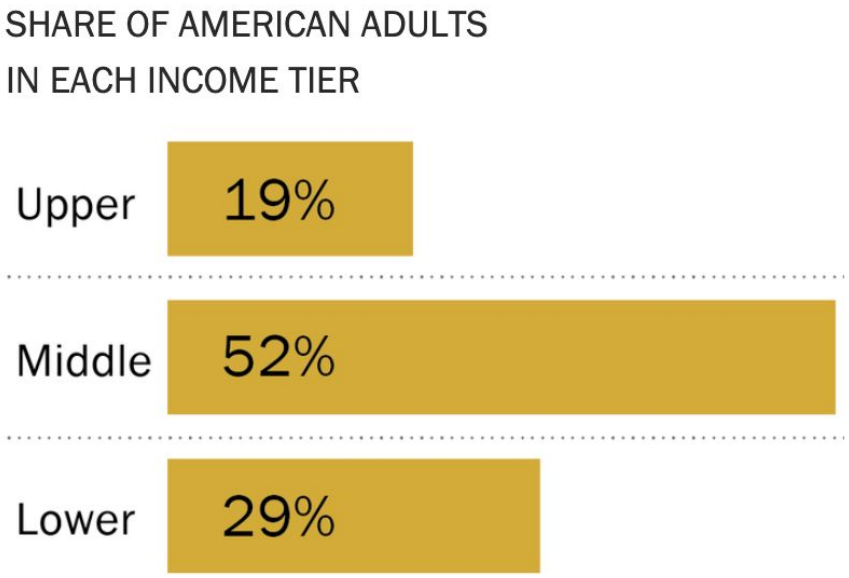

Does this chart show a distribution?

Yes!

- This chart shows the distribution of the qualitative ordinal variable "income tier."
- Each individual is in exactly one category.
- The values we see are the proportions of individuals in that category.
- Everyone is represented, as the total percentage is 100%.

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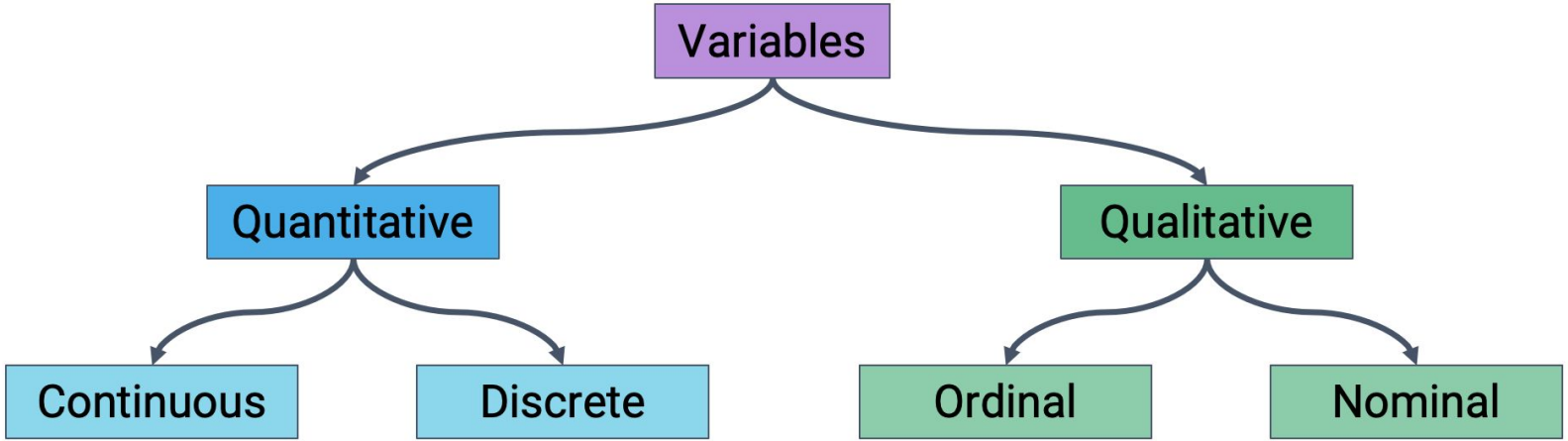
Variable types should inform plot choice

Different plots are more or less suited for displaying particular types of variables



First step of visualization: identify the variables being visualized. Then, select a plot type accordingly.

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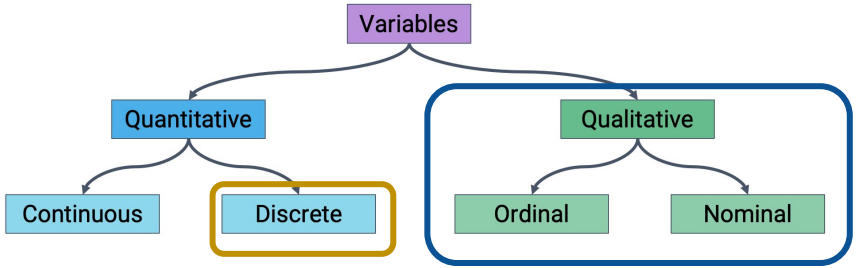
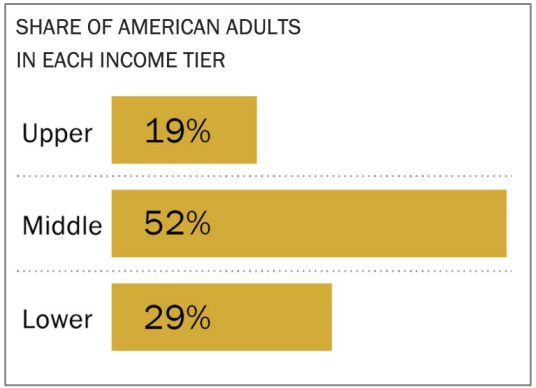
Bar plots: distributions of qualitative variables

Bar plots are the most common way of displaying the **distribution** of a **qualitative** variable.

*Sometimes quantitative discrete data too, if there are few unique values.



- For example, the proportion of adults in the upper, middle, and lower classes.
- Lengths encode values.
  - *Widths* encode *nothing*!
  - *Color* could indicate a sub-category (but not necessarily).



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World Bank Dataset

We will be using the `world_bank` dataset about world countries for most of our work today.

These data are available on MyLS!

	Continent	Country	Primary completion rate: Male: % of relevant age group: 2015	Primary completion rate: Female: % of relevant age group: 2015	Lower secondary completion rate: Male: % of relevant age group: 2015	Lower secondary completion rate: Female: % of relevant age group: 2015	Youth literacy rate: Male: % of ages 15-24: 2005-14	Youth literacy rate: Female: % of ages 15-24: 2005-14	Adult literacy rate: Male: % ages 15 and older: 2005-14	Adult literacy rate: Female: % ages 15 and older: 2005-14
0	Africa	Algeria	106.0	105.0	68.0	85.0	96.0	92.0	83.0	68.0
1	Africa	Angola	NaN	NaN	NaN	NaN	79.0	67.0	82.0	60.0
2	Africa	Benin	83.0	73.0	50.0	37.0	55.0	31.0	41.0	18.0
3	Africa	Botswana	98.0	101.0	86.0	87.0	96.0	99.0	87.0	89.0
5	Africa	Burundi	58.0	66.0	35.0	30.0	90.0	88.0	89.0	85.0

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Generating bar plots: Matplotlib

In Data 100, we will mainly use two libraries for generating plots: Matplotlib and Seaborn.

Most Matplotlib plotting functions follow the same structure: we pass in a sequence (list, array, or Series) of values to be plotted on the x-axis, and a second sequence of values to be plotted on the y-axis.

```
import matplotlib.pyplot as plt
plt.plotting_function(x_values, y_values)
```

Matplotlib is typically given the alias `plt`

To add labels and a title:

```
plt.xlabel("x axis label")
plt.ylabel("y axis label")
plt.title("Title of the plot");
```


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Generating bar plots: Matplotlib

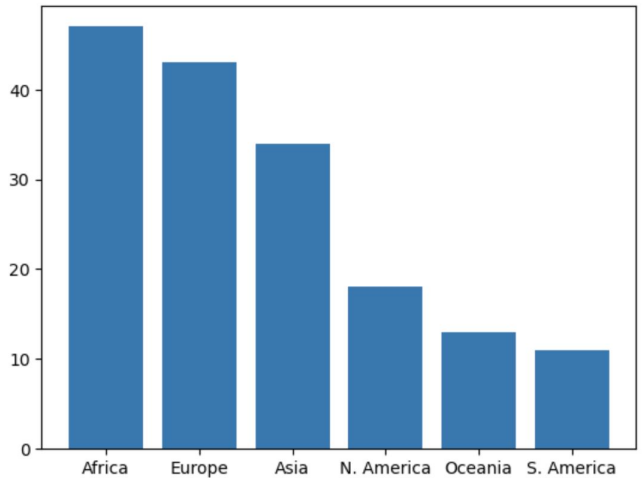
To create a bar plot in Matplotlib: `plt.bar( )`

```
continents = wb["Continent"].value_counts()
```

```
Africa        47
Europe        43
Asia          34
N. America    18
Oceania       13
S. America    11
Name: Continent, dtype: int64
```

```
plt.bar(continents.index, continents.values);
```

x values y values



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Generating bar plots: Seaborn

Seaborn plotting functions use a different structure: pass in an entire *DataFrame*, then specify what column(s) to plot.

Seaborn is typically given the alias `sns`

```
import seaborn as sns
sns.plotting_function(data=df, x="x_col", y="y_col")
```

To add labels and a title, use the same syntax as before:

```
plt.xlabel("x axis label")
plt.ylabel("y axis label")
plt.title("Title of the plot");
```


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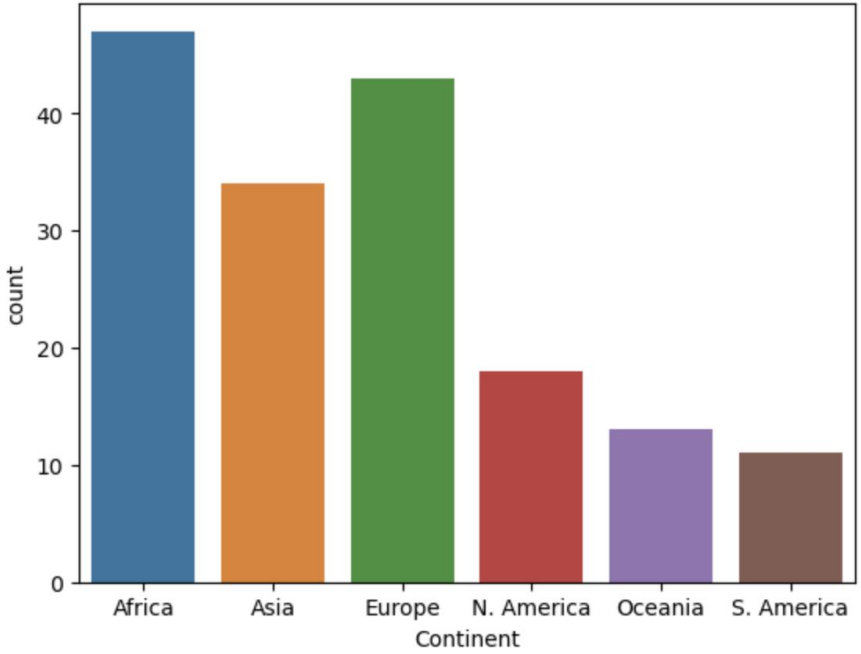
Generating bar plots: Seaborn

To create a bar plot in Seaborn: `sns.countplot( )`



`countplot` operates at a higher level of abstraction!

You give it the entire DataFrame and it does the counting for you.

```
import seaborn as sns

sns.countplot(data = wb, x = 'Continent');
```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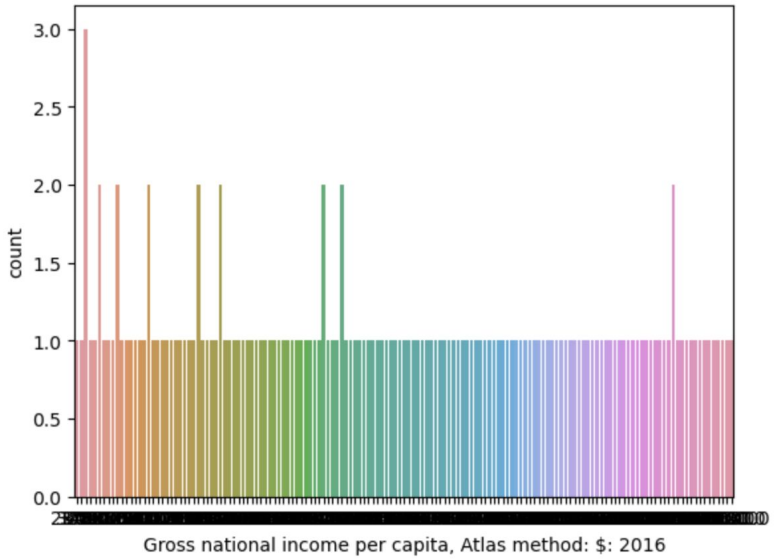
Distributions of quantitative variables

Earlier, we said that bar plots are appropriate for distributions of qualitative variables.

Why only qualitative? Why not quantitative as well?

- For example: the distribution of gross national income per capita



A bar plot will create a separate bar for each unique value. This leads to too many bars for continuous data!

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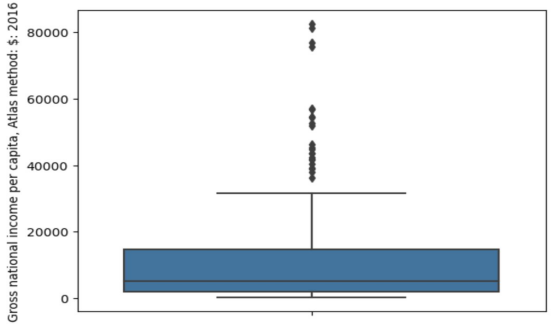
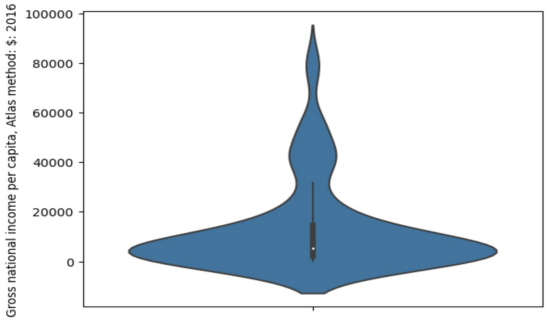
Distributions of quantitative variables

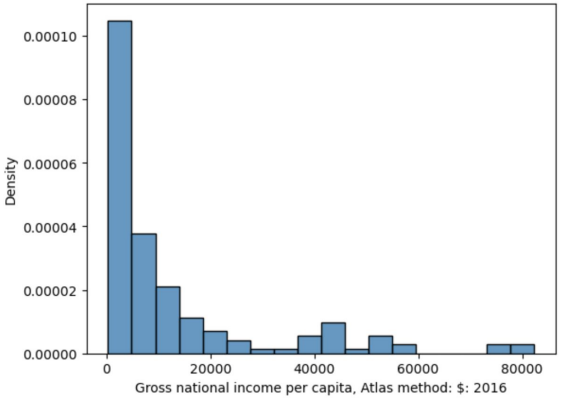
To visualize the distribution of a continuous quantitative variable:



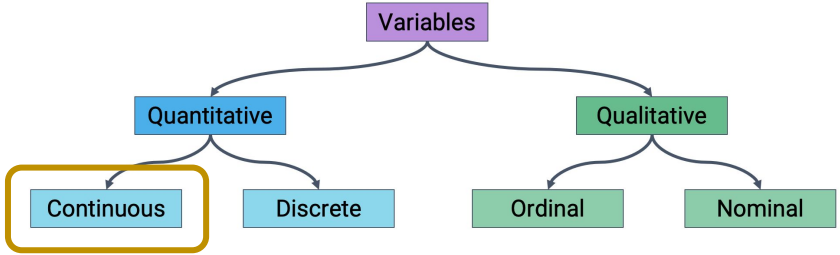
Box plot



Violin plot



Histogram



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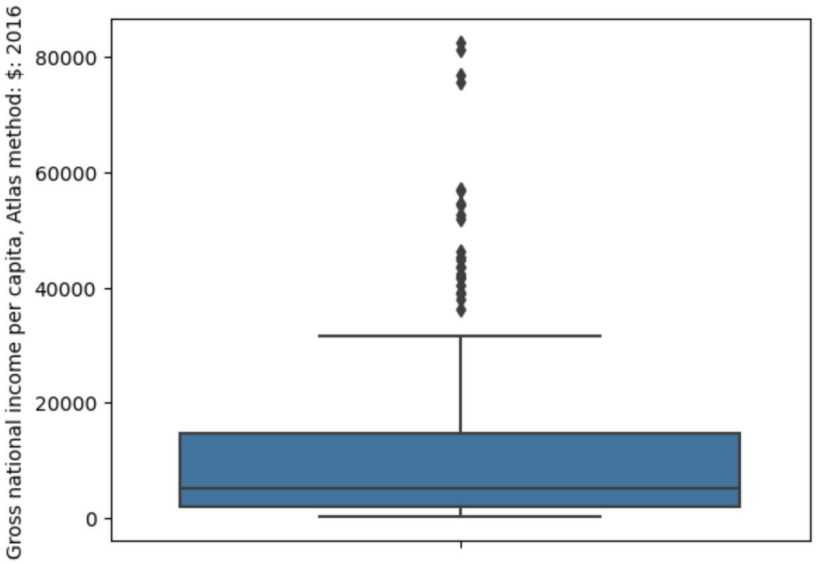
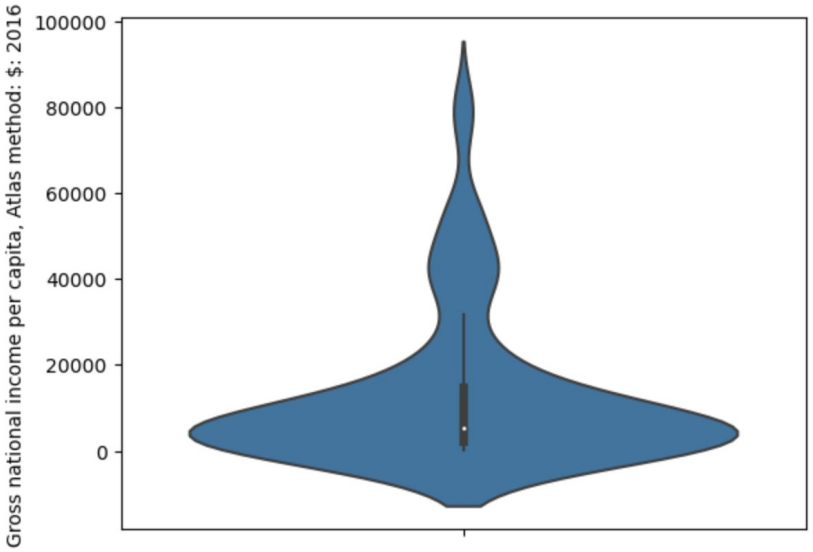
Box plots and violin plots

Box plots and violin plots display distributions using information about **quartiles**.

- In a box plot, the width of the box encodes no meaning
- In a violin plot, the width of the "violin" indicates the density of datapoints at each value






```
sns.boxplot(data = df, y = "y_variable");
```

```
sns.violinplot(data = df, y = "y_variable");
```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Quartiles

For a quantitative variable:

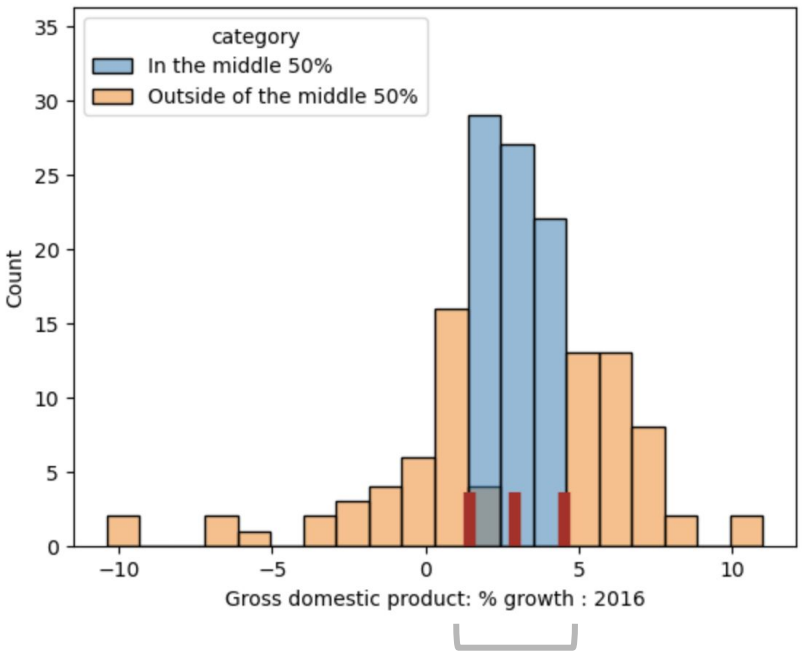
- First or lower quartile: 25th percentile
- Second quartile: 50th percentile (median)
- Third or upper quartile: 75th percentile

The interval [first quartile, third quartile] contains the "middle 50%" of the data.

Interquartile range (IQR) measures spread.

- IQR = third quartile – first quartile.

In pandas: try browsing the [reference manual](#)! (Hint: there is no IQR, you'll need a different function.)



The length of this region is the IQR

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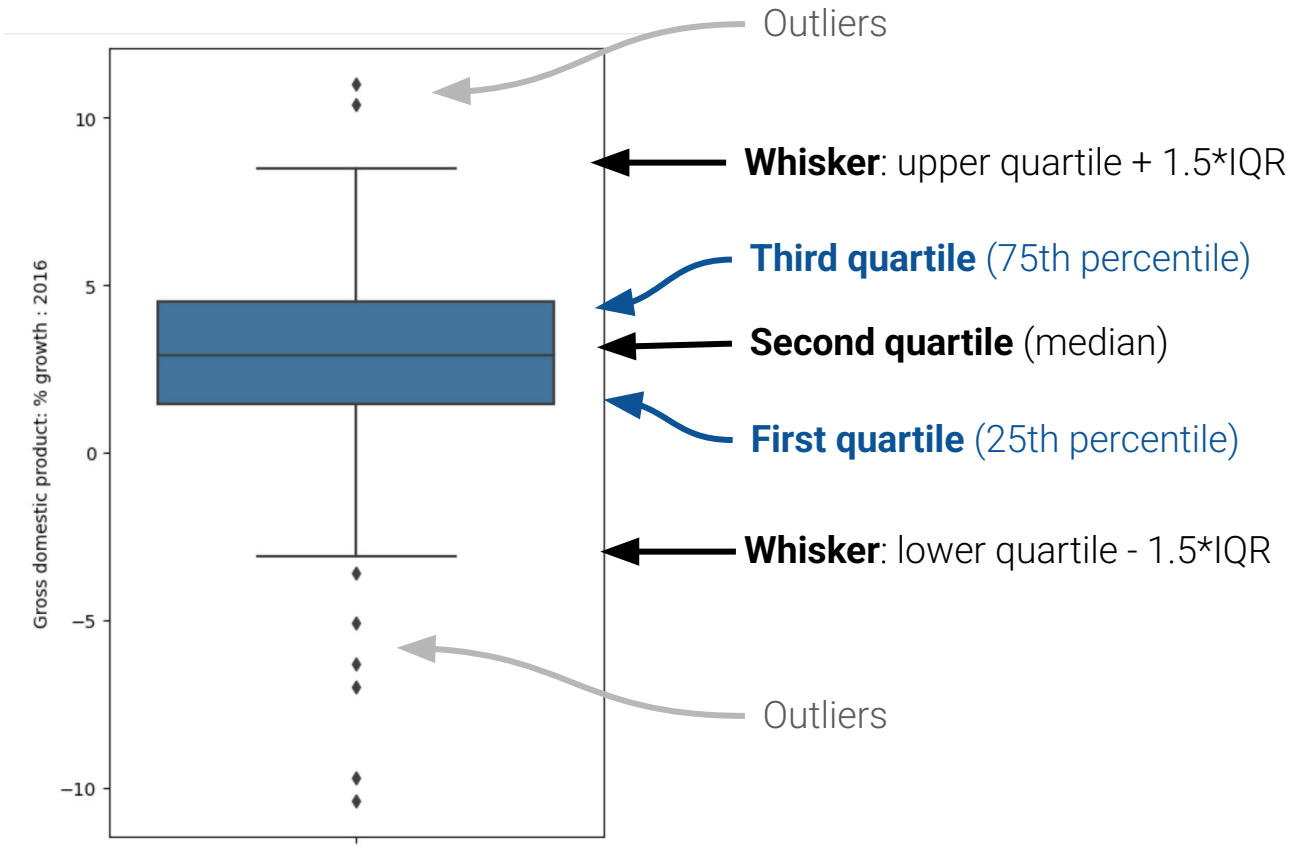
Box Plots




```
sns.boxplot(data = wb, y = 'Gross domestic product: % growth : 2016')
```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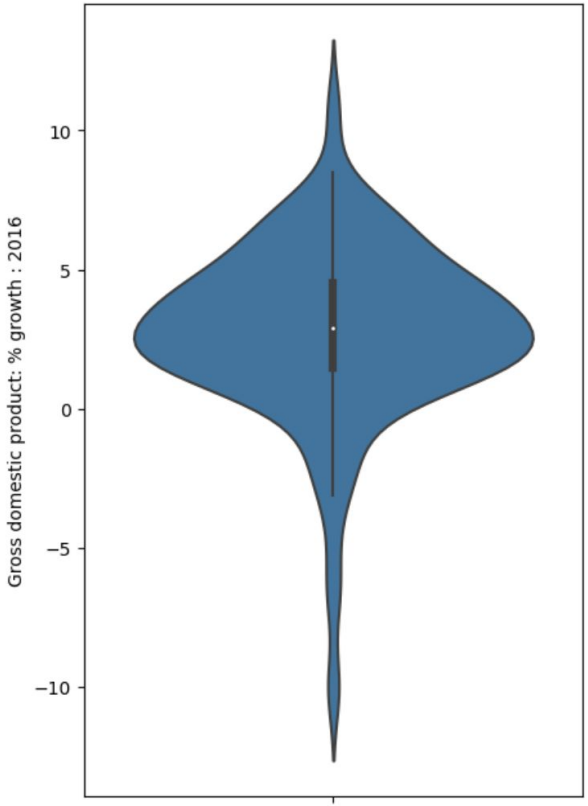
Violin Plots



Violin plots are similar to box plots, but also show smoothed density curves.

- The "width" of our "box" now has meaning!
- The three quartiles and "whiskers" are still present – look closely.

```
sns.violinplot(data = wb, y = 'Gross domestic product: % growth : 2016')
```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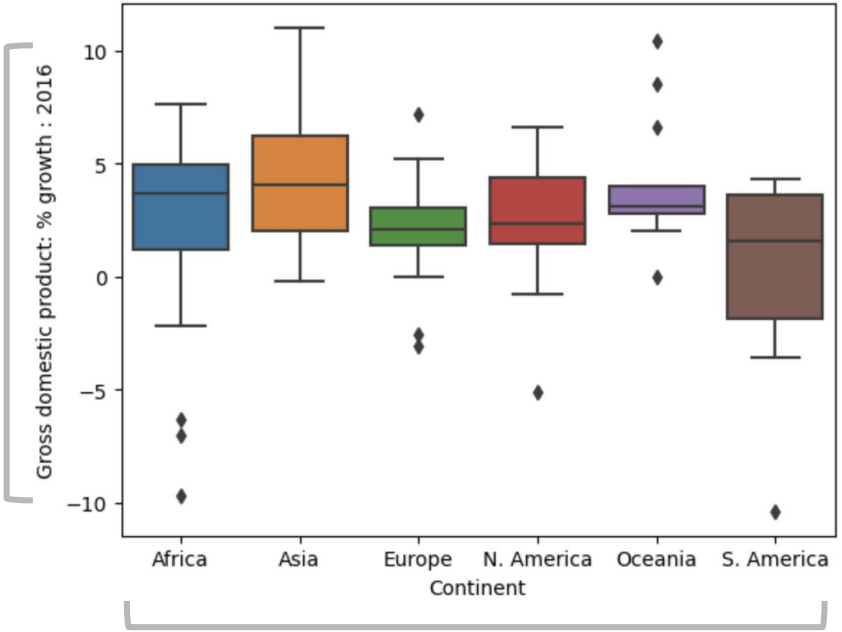
Side-by-side box and violin plots

What if we wanted to incorporate a *qualitative* variable as well? For example, compare the distribution of a quantitative continuous variable *across* different qualitative categories.

```
sns.boxplot(data=wb, x= “Continent”, y= “Gross domestic product: % growth : 2016”);
```

GDP growth:
quantitative continuous



Continent: qualitative nominal

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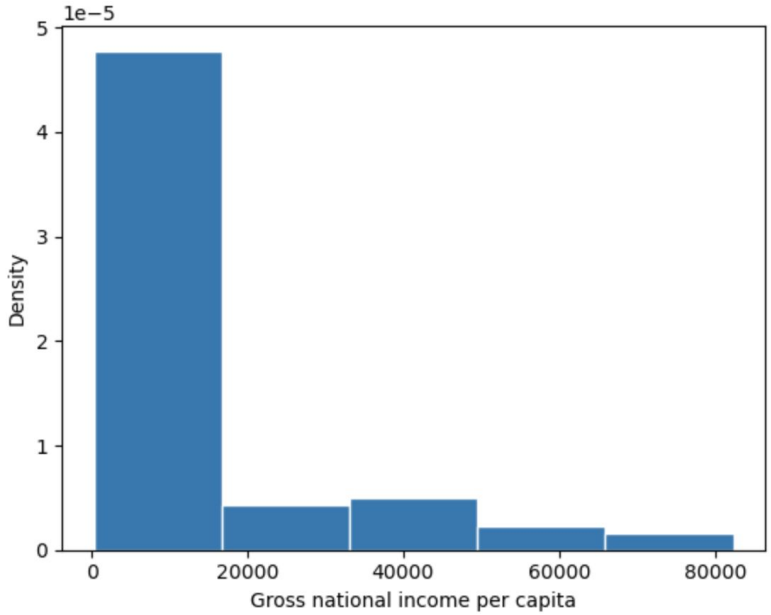
Histograms

A histogram:

- Collects datapoints with similar values into a shared “bin”
- Scales the bins such that the **area** of each bin is equal to the **percentage** of datapoints it contains (as in Data 8)



The first bin has a width of $16410
height of 4.77×10^5

This means that it contains $16410 \times (4.77 \times 10^5) = 78.3\%$ of all datapoints in the dataset

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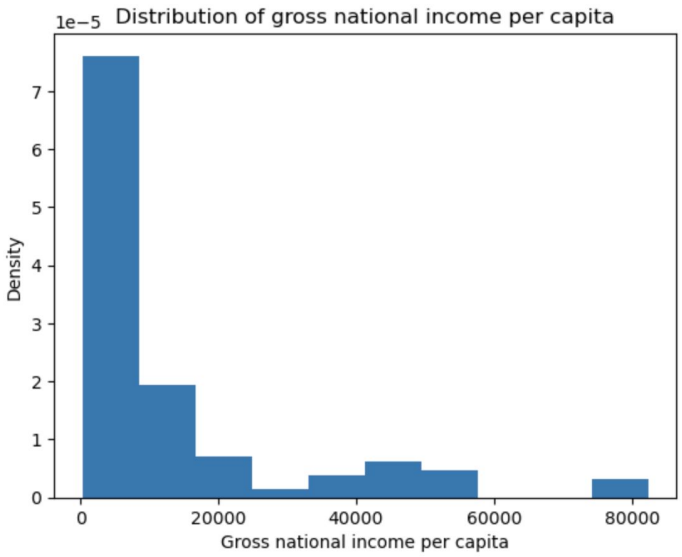
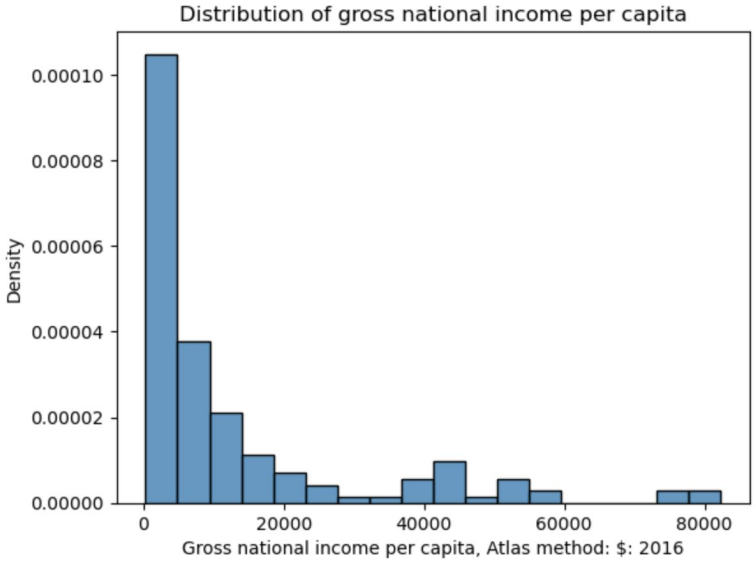
Histograms in code

In Matplotlib: `plt.hist(x_values, density = True)`

In Seaborn: `sns.histplot(data = df, x = “x_column”, stat = “density”)`



Matplotlib



Seaborn

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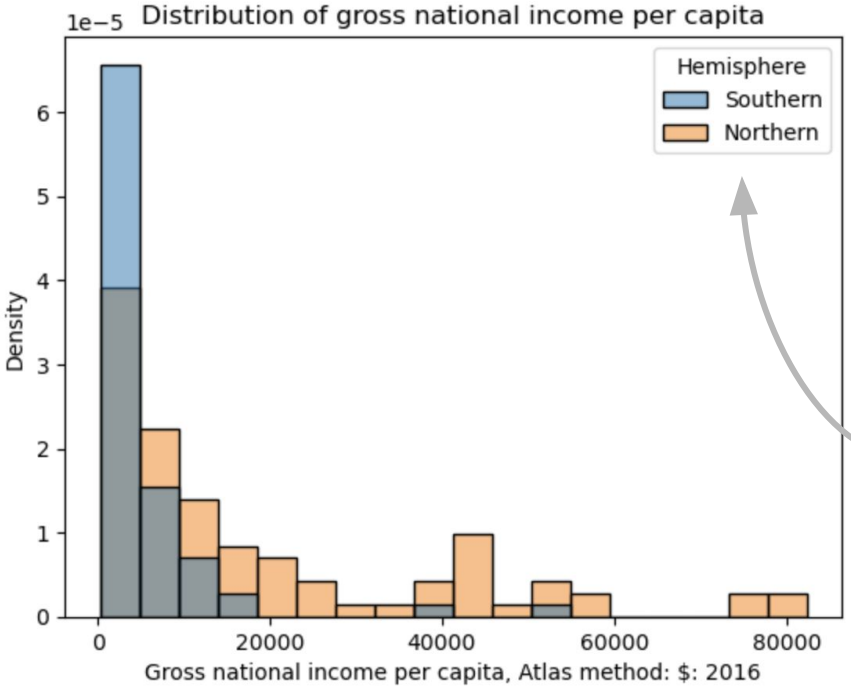
Overlaid histograms

To compare a quantitative variable's distribution across qualitative categories, overlay histograms on top of one another.



The **hue** parameter of Seaborn plotting functions sets the column that should be used to determine color.

```
sns.histplot(data = wb, hue="Hemisphere",
x="Gross national income...")
```

Always include a legend when color is used to encode information!

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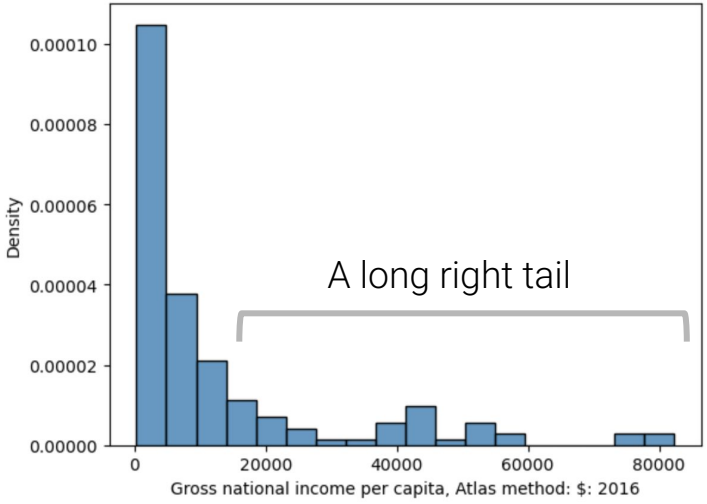
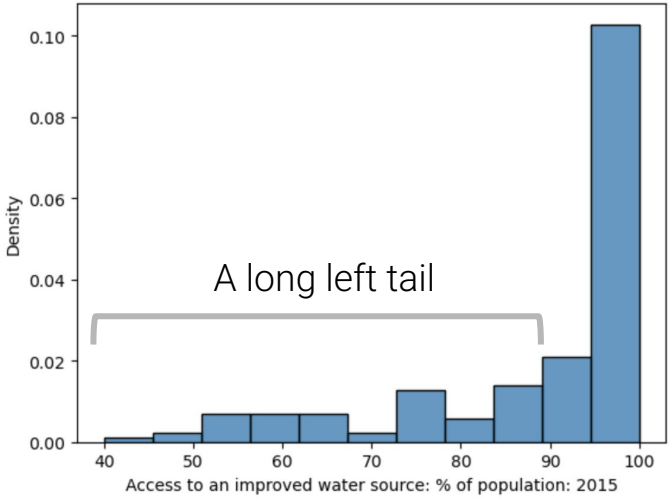
Interpreting histograms

The **skew** of a histogram describes the direction in which its “tail” extends

- A distribution with a long right tail is skewed right
- A distribution with a long left tail is skewed left

A histogram with no clear skew is called symmetric.





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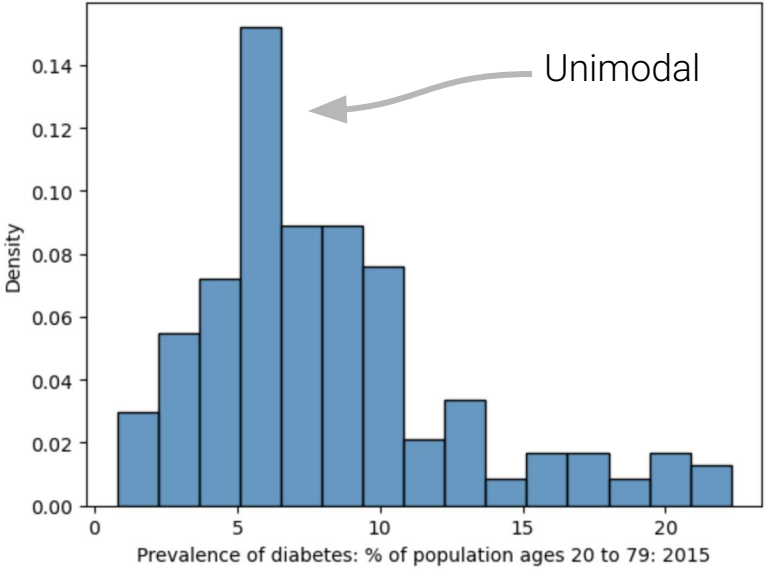
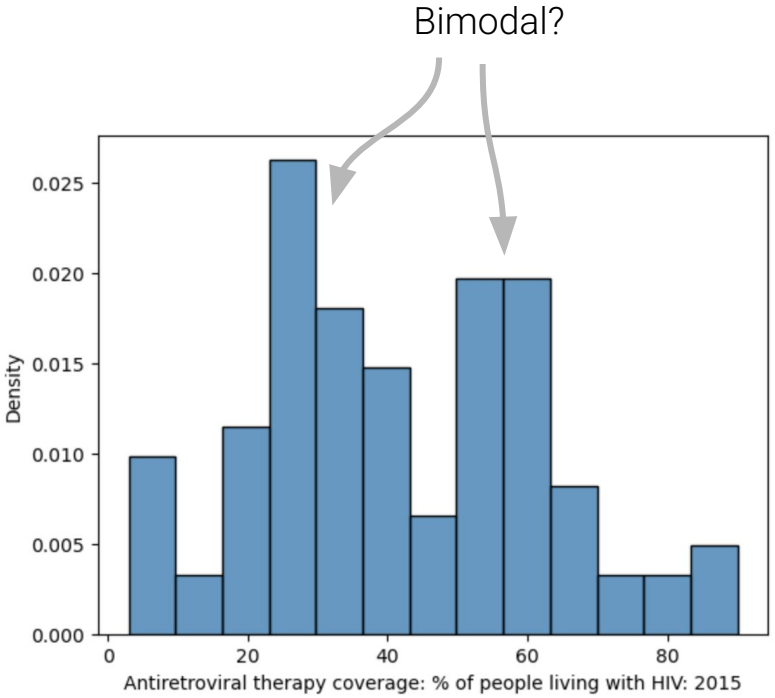
Interpreting histograms

The **mode(s)** of a histogram are the peak values in the distribution

- A distribution with one clear peak is called unimodal
- Two peaks: bimodal
- More peaks: multimodal





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Kernel Density Estimation

- Goals of visualization
- Visualizing distributions
- **Kernel density estimation**
- Visualizing relationships
- Transformations

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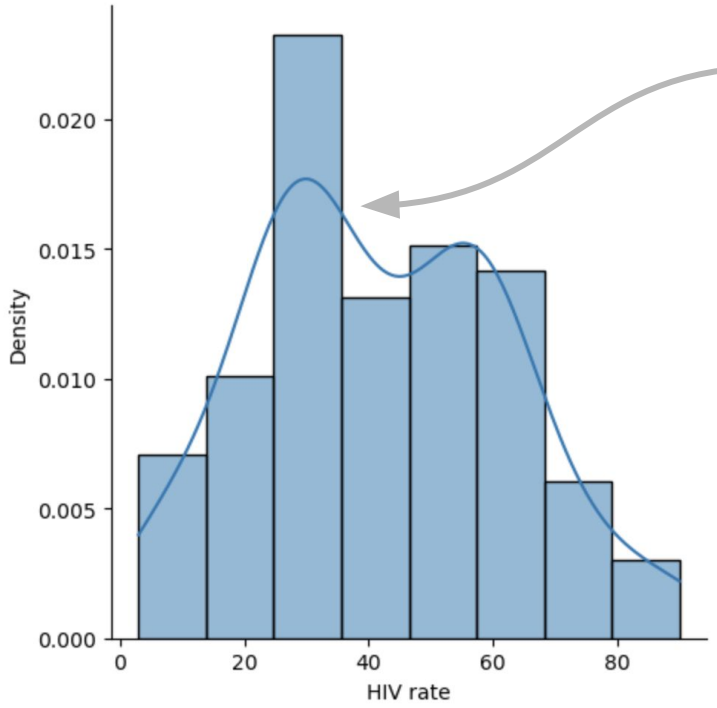
Kernel density estimation: intuition

Often, we want to identify *general* trends across a distribution, rather than focus on detail. Smoothing a distribution helps generalize the structure of the data and eliminate noise.



A KDE curve

Idea: approximate the probability distribution that generated the data

- Assign an “error range” to each data point in the dataset – if we were to sample the data again, we might get a different value
- Sum up the error ranges of all data points
- Scale the resulting distribution to integrate to 1

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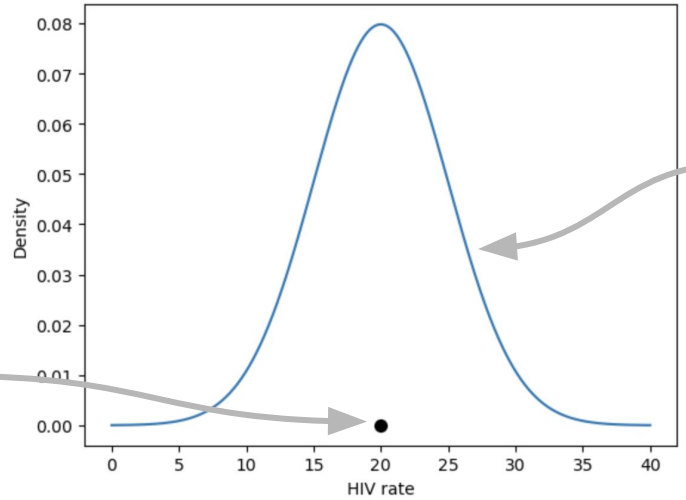
Kernel density estimation: process

Idea: approximate the probability distribution that generated the data

- Place a kernel at each data point.
- Normalize kernels so that total area = 1.
- Sum all kernels together.

A **kernel** is a function that tries to capture the randomness of our sampled data.



The **kernel** models the probability of us sampling that datapoint

Area below integrates to 1

A **datapoint** in our dataset

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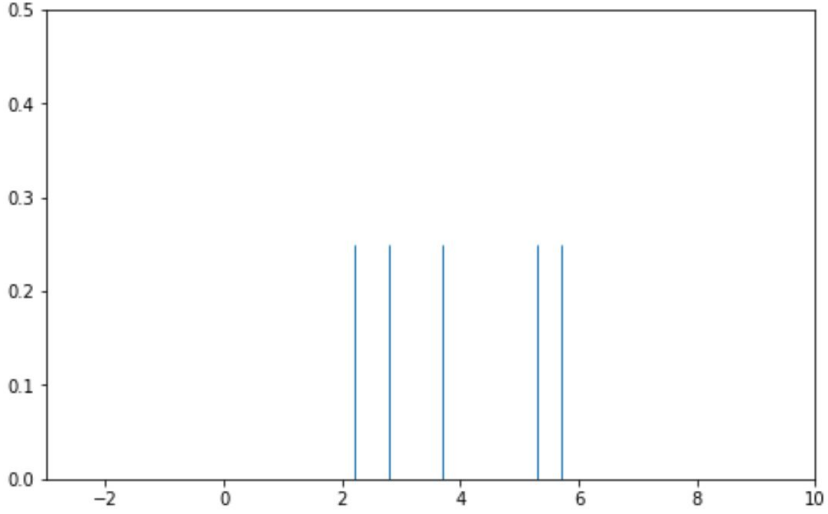
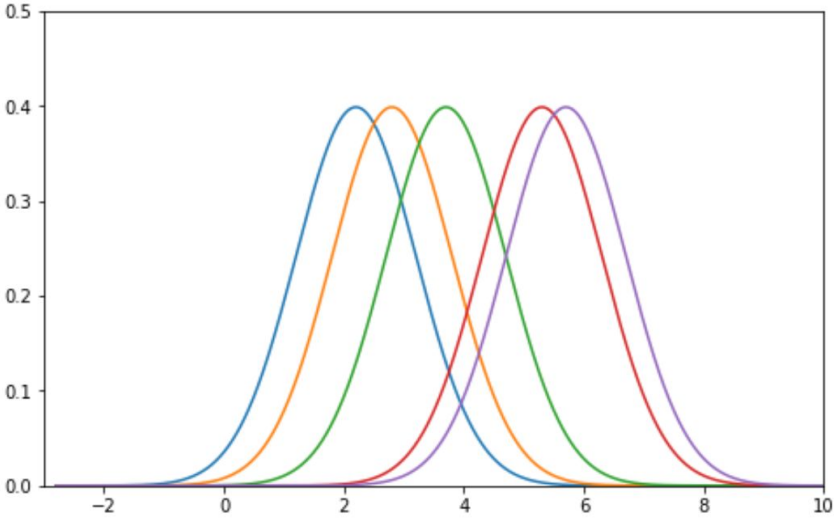
Step 1 – Place a kernel at each data point

Consider a fake dataset with just five collected datapoints.

- Place a **Gaussian kernel** with **bandwidth** of **alpha = 1**.
- We will precisely define both the **Gaussian kernel** and **bandwidth** in a few slides.



Each line represents a datapoint in the dataset (e.g. one country's HIV rate)



Place a kernel on top of each datapoint

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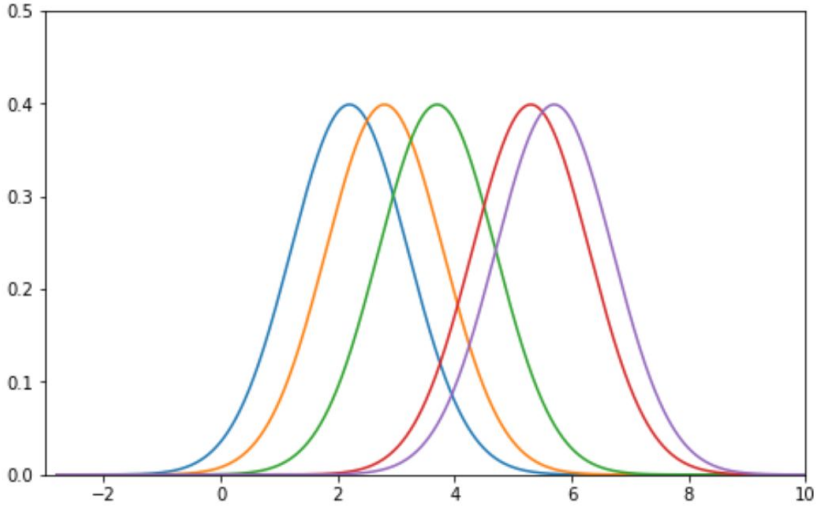
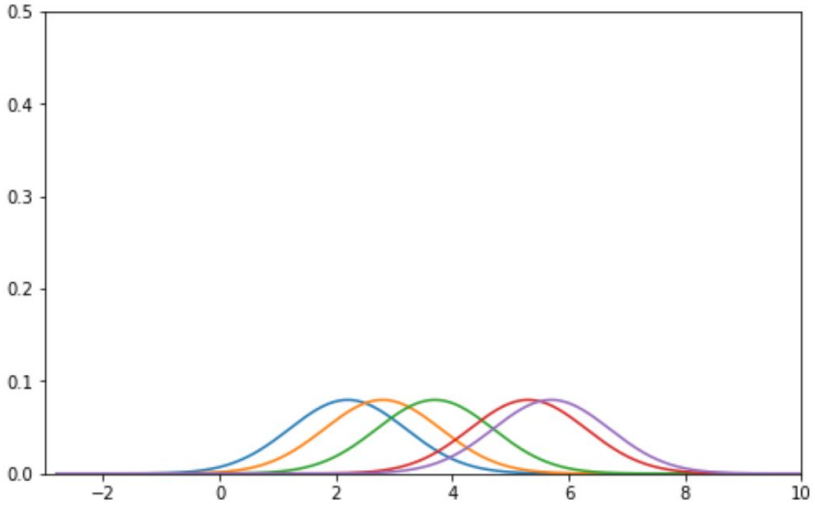
Step 2 – Normalize kernels

In Step 3, we will be summing each of these kernels to produce a probability distribution.

- We want the result to be a valid probability distribution that has area 1.
- We have 5 different kernels, each with an area 1.
- So, we normalize by multiplying each kernel by 1/5.



Each kernel has area 1



Each normalized kernel has density 1/5

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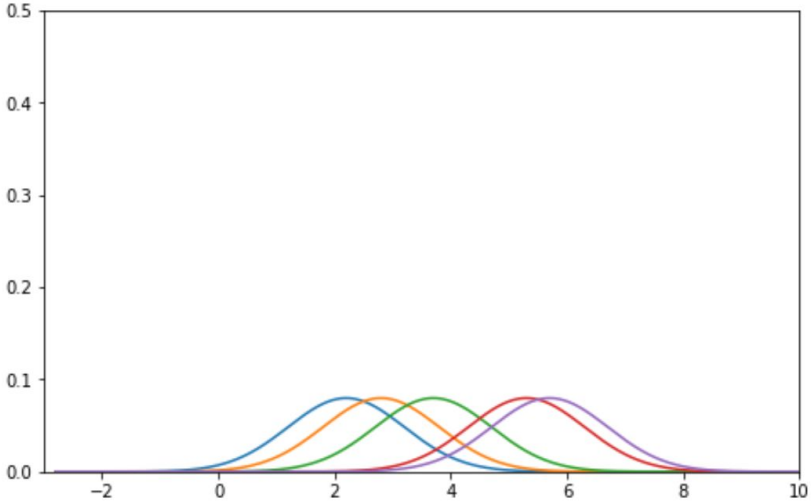
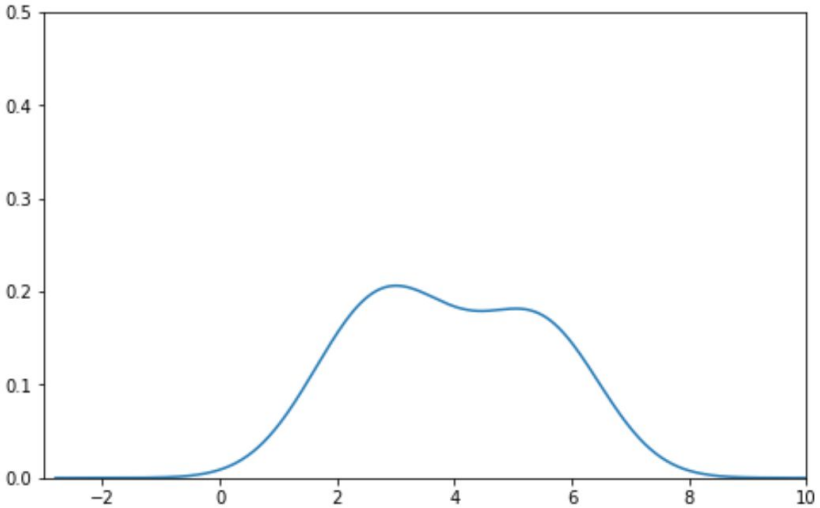
Step 3 – Sum the normalized kernels

At each point in the distribution, add up the values of all kernels. This gives us a smooth curve with area 1 – an approximation of a probability distribution!



Sum these five normalized curves together



The final KDE curve

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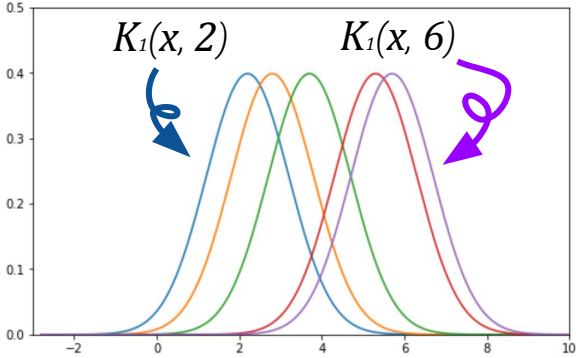
Summary of KDE

$$f_\alpha(x) = \frac{1}{n} \sum_{i=1}^{n} K_\alpha(x, x_i)$$

2 3 1



A general "KDE formula" function is given above.

- $K_\alpha(x, x_i)$ is the **kernel** function centered on the observation i.
  - Each kernel individually has area 1.
  - K represents our kernel function of choice. We'll talk about the math of these functions soon.

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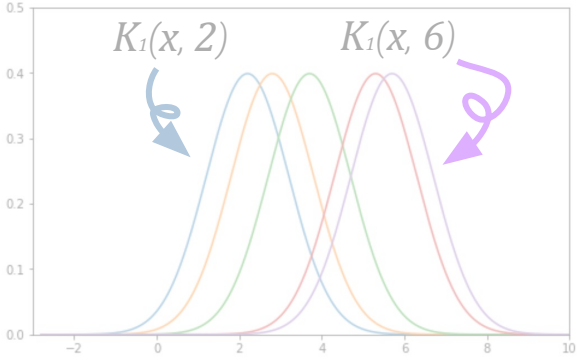
Summary of KDE

$$f_\alpha(x) = \frac{1}{n}\sum_{i=1}^{n} K_\alpha(x, x_i)$$



A general "KDE formula" function is given above.

1. $K_\alpha(x, x_i)$ is the **kernel** centered on the observation i.
   - Each kernel individually has area 1.
   - x represents any number on the number line. It is the input to our function.
2. n is the number of observed data points that we have.
   - We multiply by $1/n$ to normalize the kernels so that the total area of the KDE is still 1.
3. Each x_i (x_1, x_2, ..., x_n) represents an observed data point. We sum the kernels for each datapoint to create the final KDE curve

α is the **bandwidth** or **smoothing parameter**.

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Kernels

A **kernel** (for our purposes) is a valid density function, meaning:

- It must be non-negative for all inputs.
- It must integrate to 1(area under curve = 1).

The most common kernel is the **Gaussian kernel**.

- Gaussian = normal distribution = bell curve
- Here, x represents any input, and x_i represents the ith observed value (datapoint).
- Each kernel is **centered** on our observed values (and so its distribution mean is x_i).
- **α** is the **bandwidth parameter**. It controls the smoothness of our KDE. Here, it is also the standard deviation of the Gaussian.

$$K_\alpha(x, x_i) = \frac{1}{\sqrt{2\pi\alpha^2}} e^{-\frac{(x-x_i)^2}{2\alpha^2}}$$

Memorizing this formula is less important than knowing the shape and how the bandwidth parameter **α** smoothes the KDE.

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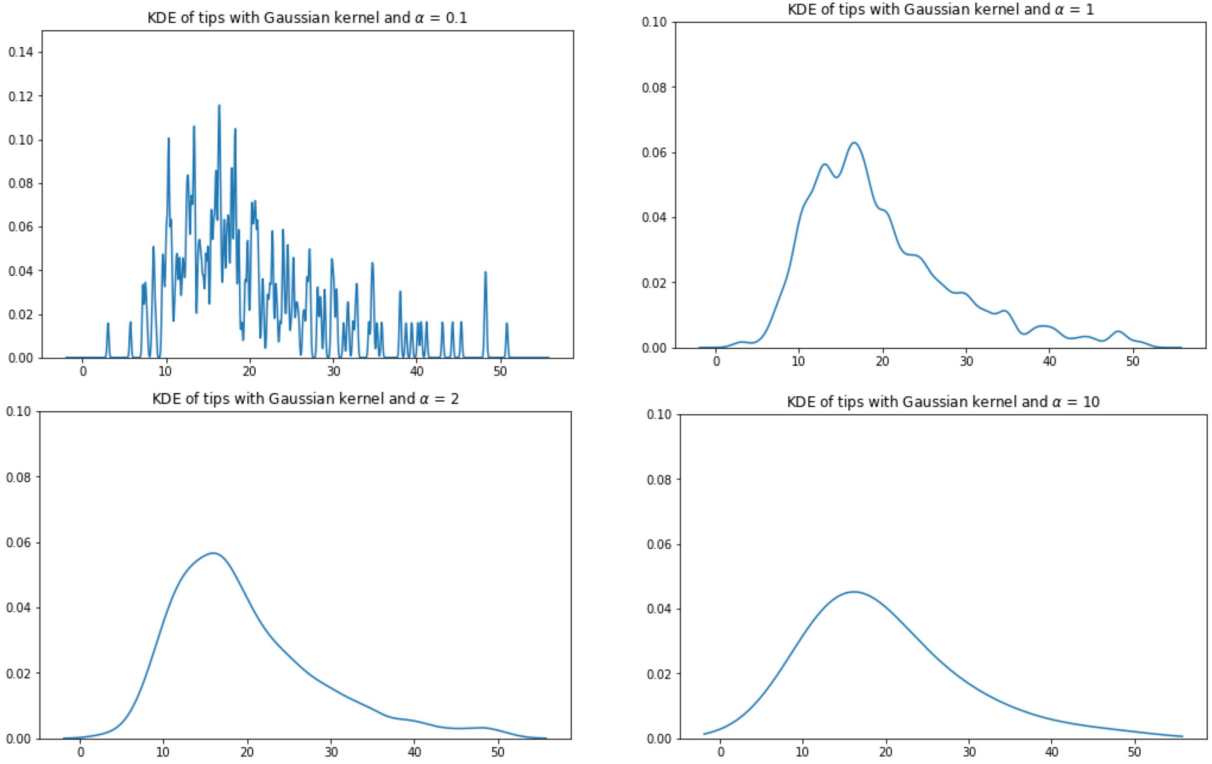
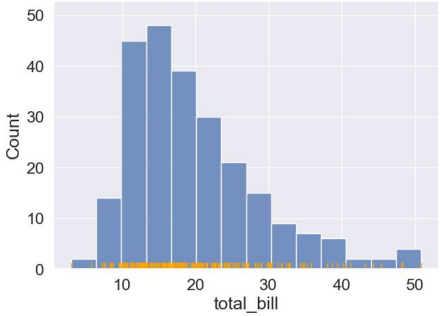
Effect of bandwidth on KDEs



Bandwidth is analogous to the width of each bin in a histogram.

- As **α** increases, the KDE becomes more smooth.
- Large **α** KDE is simpler to understand, but gets rid of potentially important distributional information (e.g. multimodality).



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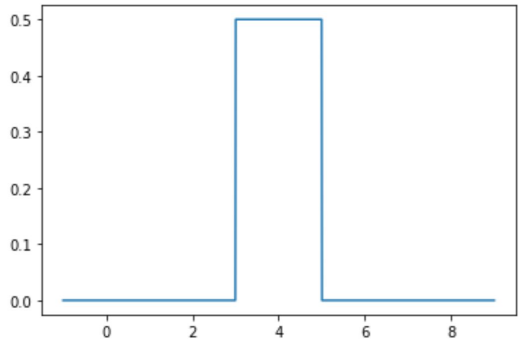
Other kernels: boxcar

As an example of another kernel, consider the **boxcar kernel**.

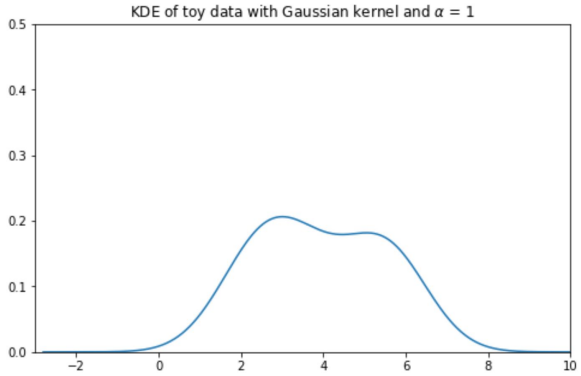
- It assigns uniform density to points within a "window" of the observation, and 0 elsewhere.
- Resembles a histogram… *sort of*.

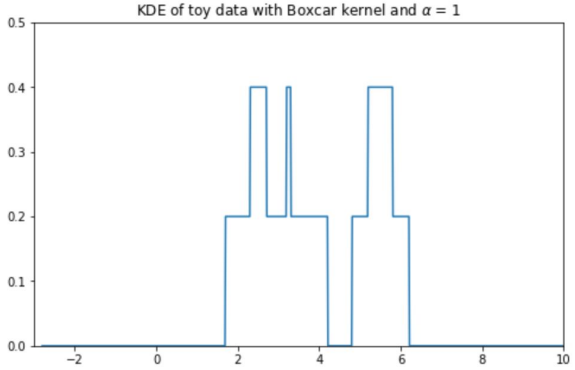
$$K_\alpha(x, x_i) = \begin{cases} \frac{1}{\alpha}, & |x - x_i| \leq \frac{\alpha}{2} \\ 0, & \text{else} \end{cases}$$

- Not of any practical use in Data 100! Presented as a simple theoretical alternative.



A boxcar kernel centered on $x_i = 4$ with $\boldsymbol{\alpha} = 2$.





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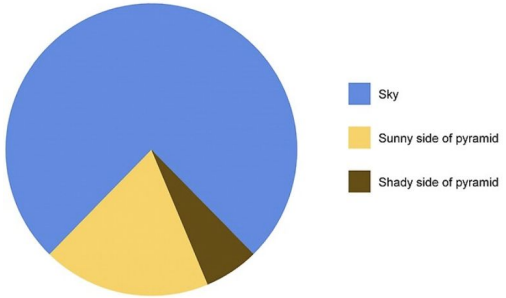
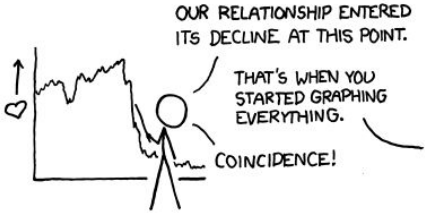
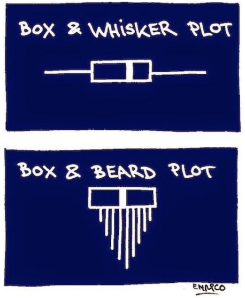
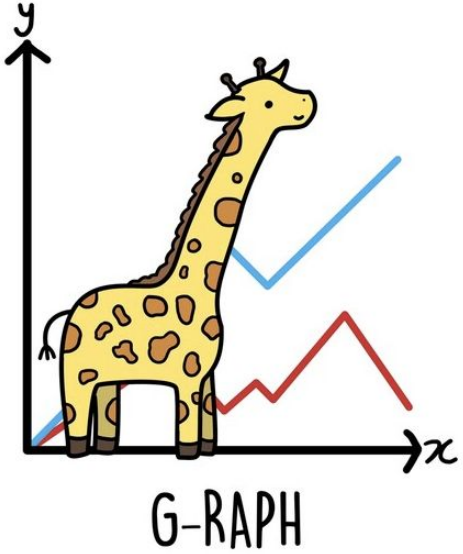
Interlude









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Visualizing Relationships

- Goals of visualization
- Visualizing distributions
- Kernel density estimation
- **Visualizing relationships**
- Transformations

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From distributions to relationships

Up until now, we focused exclusively on visualizing variable distributions.

Now we will visualize **relationships** between variables. In other words, how do sets of two (or more) variables vary in relation to one another?

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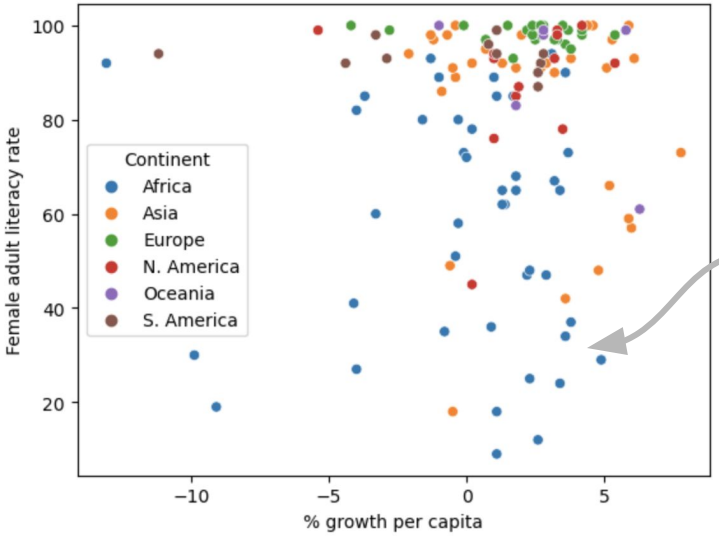
Scatter plot

Scatter plots are used to reveal relationships between two quantitative variables

- Plot one quantitative continuous variable on the x-axis, and second quantitative continuous variable on the y-axis
- Each scatter point represents one datapoint in the dataset




```
plt.scatter(x_values, y_values)
```

```
sns.scatterplot(data = df, x = “x_column”, \
                y = “y_column”, hue = “hue_column”)
```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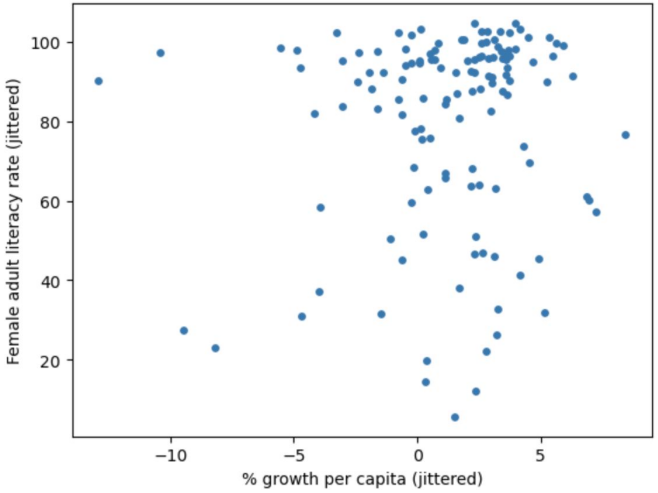
Overplotting

The plot on the previous slide suffered from **overplotting** – scatter points all stacked on top of one another are difficult to see.

Jittering: adding a small amount of random noise to all x and y values to slightly move each scatter point. Main trends are still present, but individual datapoints are easier to distinguish.




```
x_noise = np.random.uniform(-1, 1, len(wb))
y_noise = np.random.uniform(-5, 5, len(wb))

plt.scatter(wb['% growth'] + x_noise, \
            wb['Literacy rate: Female'] + y_noise, \
            s=15);
```

Decreasing point size also helps. `s` specifies the marker size in Matplotlib.

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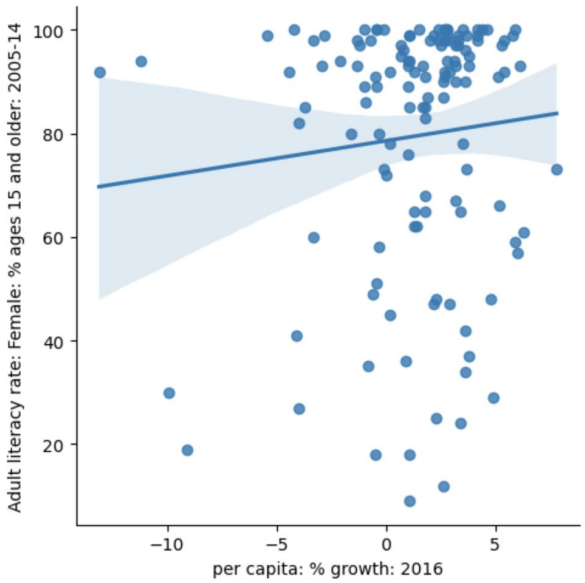
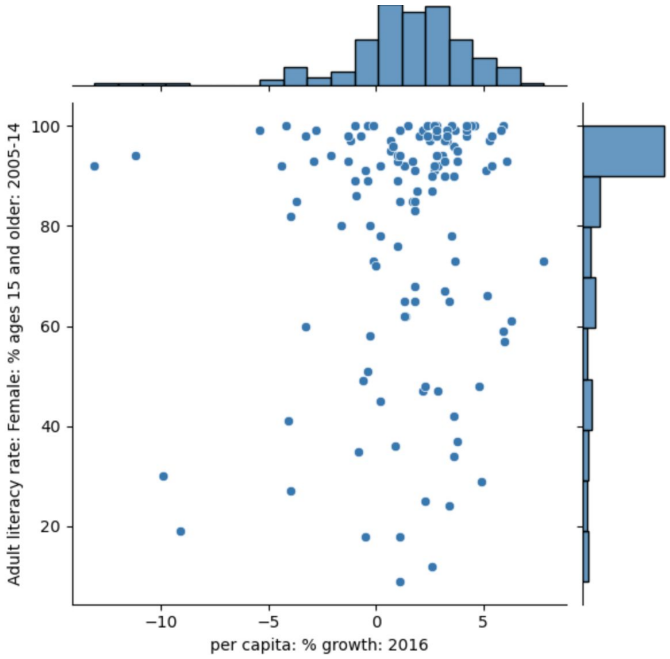
Scatter plot alternatives

Seaborn includes several built-in functions for making more complex scatter plots.






```
sns.lmplot(data = df, \
x = “x_column”, y = “y_column”)
```

```
sns.jointplot(data = df, \
x = “x_column”, y = “y_column”)
```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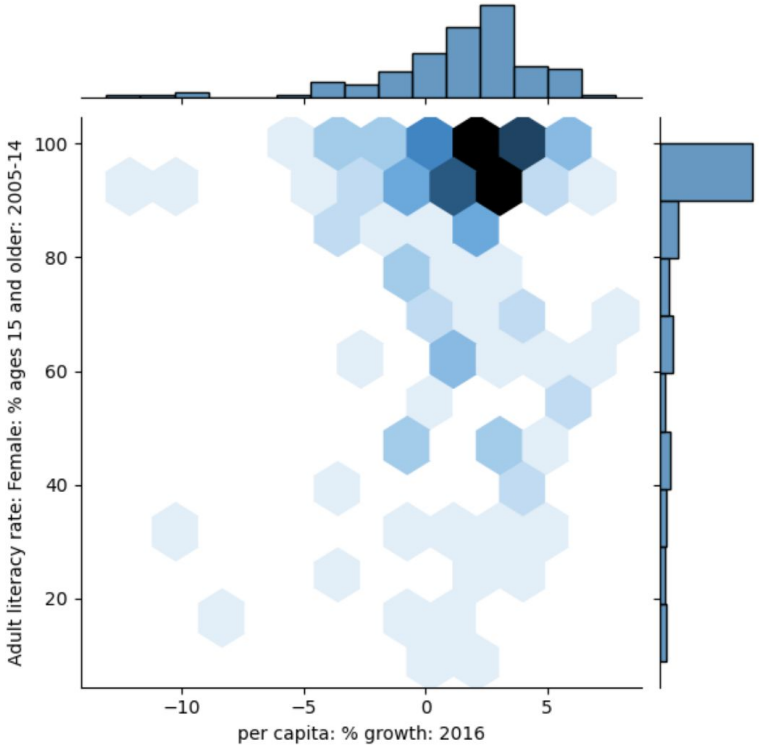
Hex plots

Rather than plot individual datapoints, plot the *density* of their joint distribution.

Can be thought of as a two dimensional histogram.

- The xy plane is binned into hexagons.
- More shaded hexagons typically indicate a greater density/frequency = more datapoints lie in that spot

```
sns.jointplot(data=df, x='x_column', \
y='y_column', kind='hex')
```



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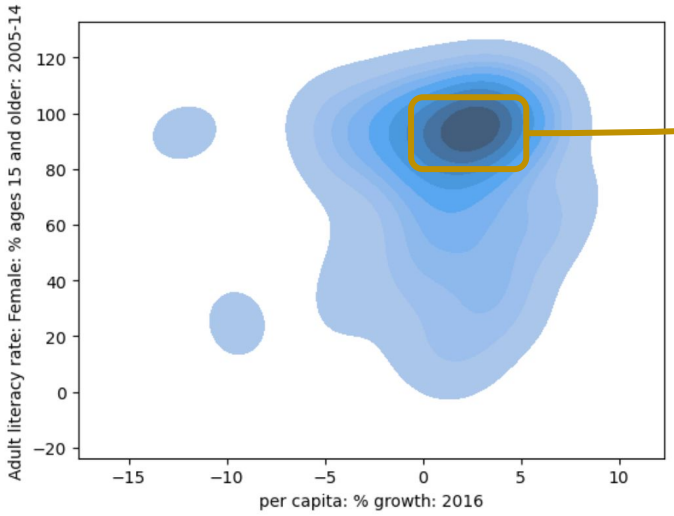
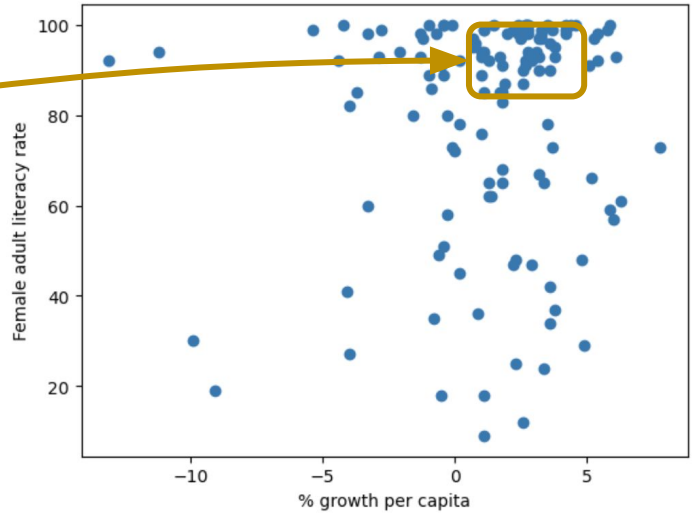
Contour plots

2-dimensional version of a KDE plot

Similar to a topographic map – contour lines represent an area that has the *same density* of datapoints throughout. Darker colors indicate more datapoints in the region.



Dark color → many datapoints




```
sns.kdeplot(data = df, x='x_column', y='y_column', fill=True)
```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Transformations

- Goals of visualization
- Visualizing distributions
- Kernel density estimation
- Visualizing relationships
- **Transformations**

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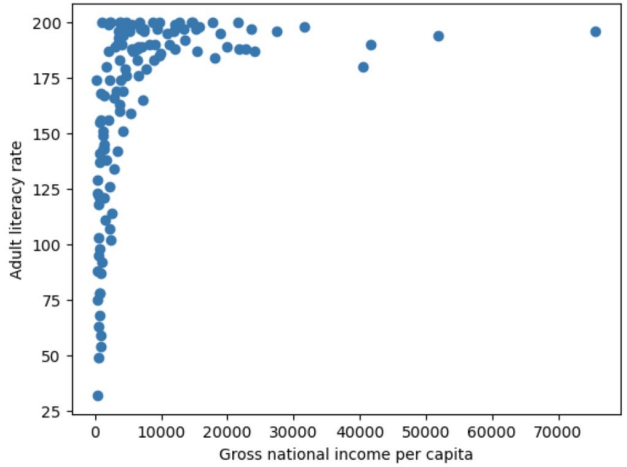
Visualization theory

Remember our goals of visualization:

1. To help your own understanding of your data/results
2. To communicate results/conclusions to others

These are influenced by our choice of visualization and our choices in *how to prepare data for visualization*.



What problems are there here?

- Data is “smushed” – hard to interpret, even if we jittered
- Difficult to generalize a clear relationship between the variables

We often **transform** a dataset to help prepare it for being visualized.

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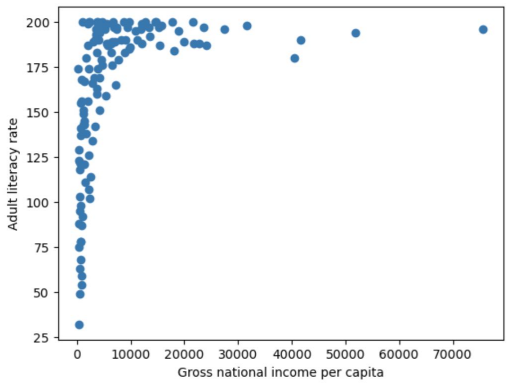
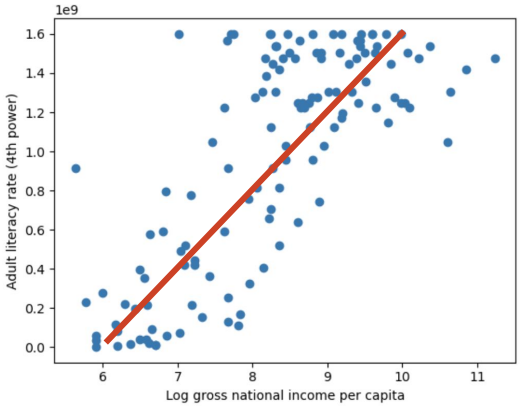
Linearization

When applying transformations, we often want to **linearize** the data – rescale the data so the x and y variables share a linear relationship.





Why?

- Linear relationships are simple to interpret – we know how to work with slopes and intercepts to understand how two variables are related
- Starting next week, we will start building linear models – these are more effective with linearized data

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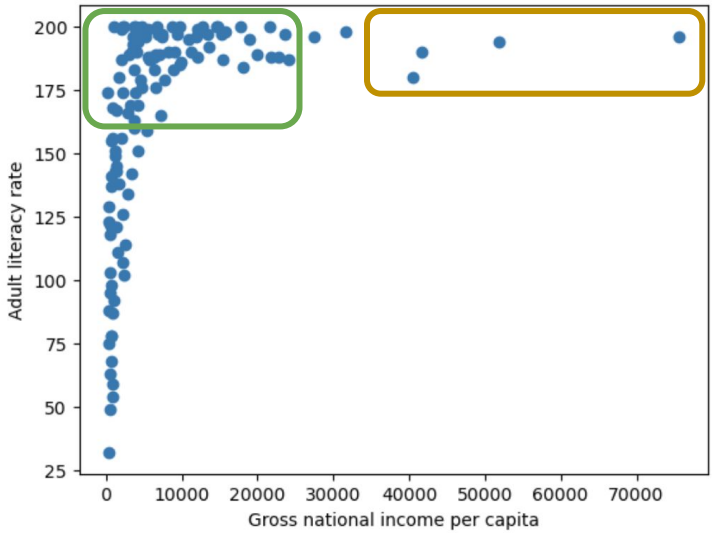
Applying transformations

What makes this plot non-linear?



1. A few large outlying x values are distorting the horizontal axis

2. Many large y values are all clumped together, compressing the vertical axis

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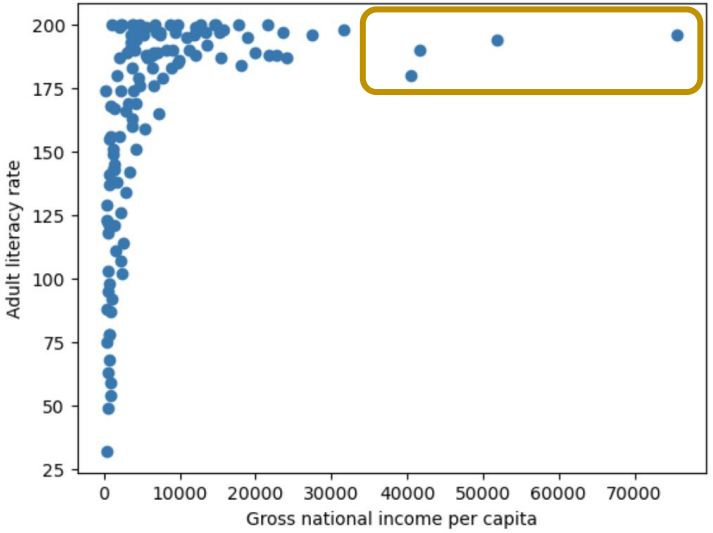
Applying transformations

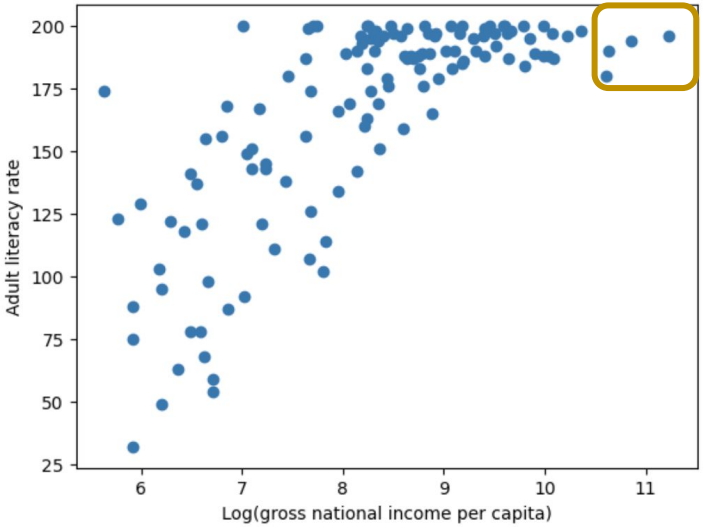
What makes this plot non-linear?

1. A few large outlying x values are distorting the horizontal axis

Resolve by log-transforming the x data:

- Taking the log of a large number decreases its value significantly
- Taking the log of a small number does not change its value as significantly





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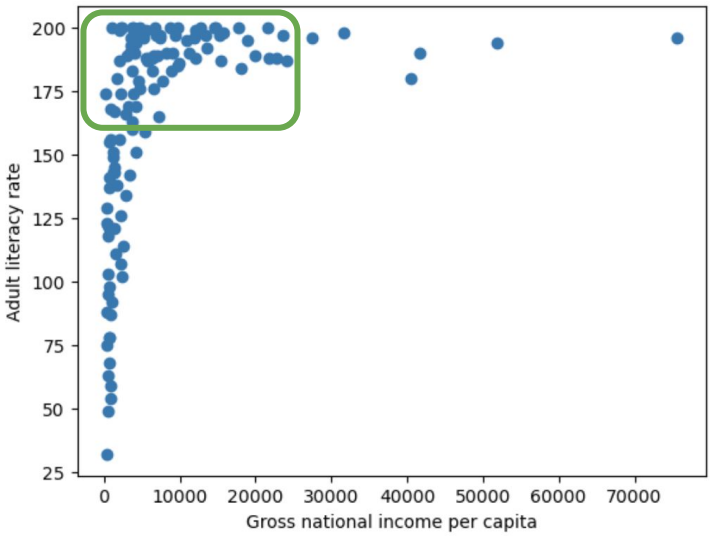
Applying transformations

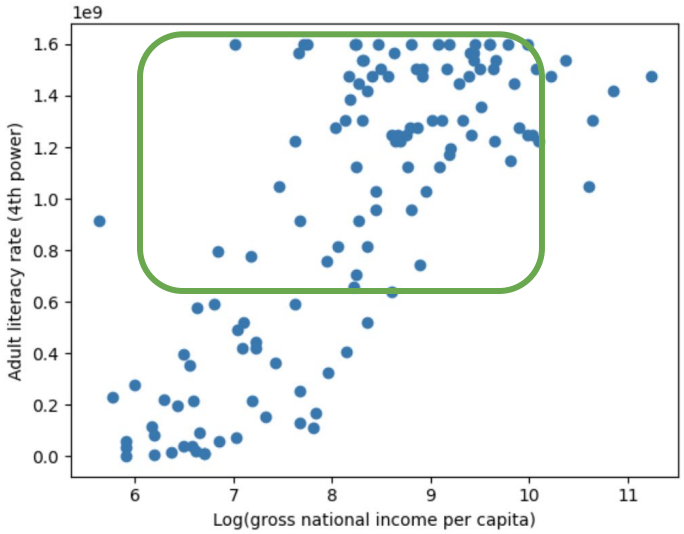
What makes this plot non-linear?

2. Many large y values are all clumped together, compressing the vertical axis

Resolve by power-transforming the x data:

- Raising a large number to a power increases its value significantly
- Raising a small number to a power does not change its value as significantly





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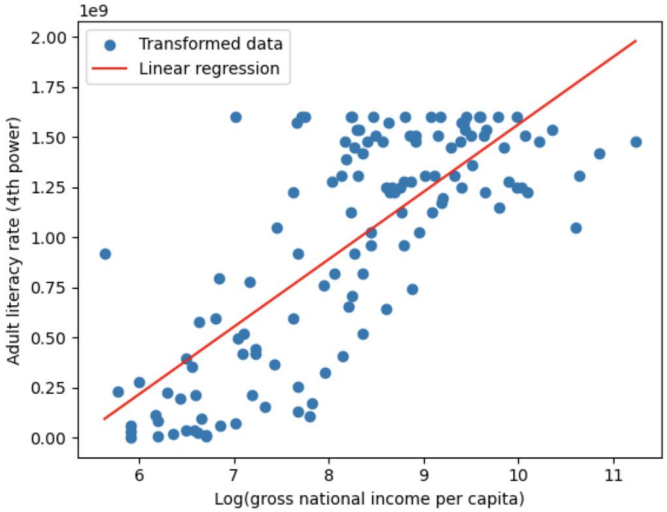
Interpreting transformed data

Now, we see a linear relationship between the transformed variables

This tells us about the underlying relationship between the *original* x and y!

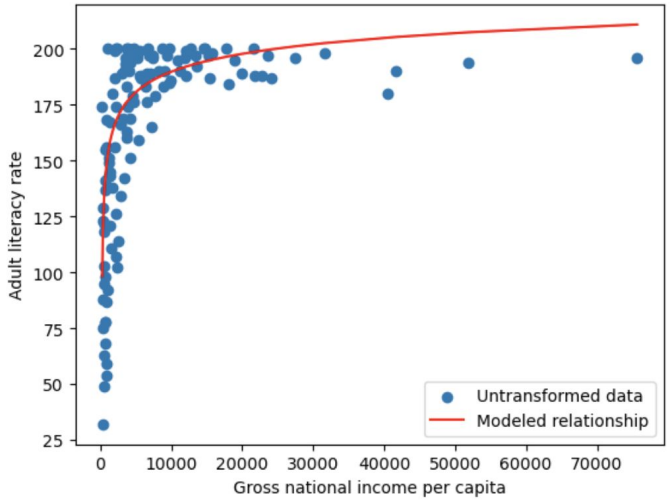
$$y^4 = m(\log x) + b \longrightarrow y = [m(\log x) + b]^{1/4}$$

y^4



$\log x$

y



x

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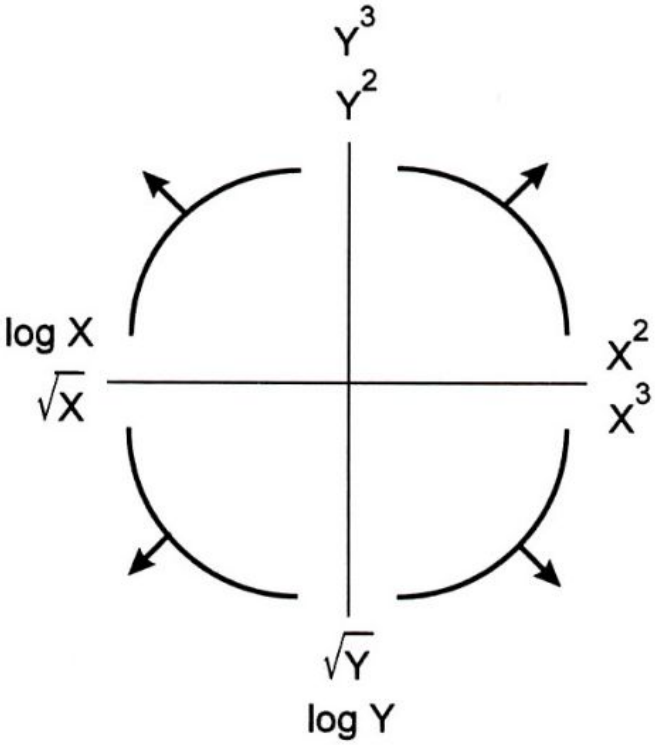
Tukey-Mosteller Bulge Diagram

The **Tukey-Mosteller Bulge Diagram** is a guide to possible transforms to try to get linearity.

- A visual summary of the reasoning we just worked through
- sqrt and log make a value “smaller”.
- Raising to a value to a power makes it “bigger”.
- There are multiple solutions. Some will fit better than others.



You should still understand the *logic* we just worked through to decide how to transform the data. The bulge diagram is just a summary.

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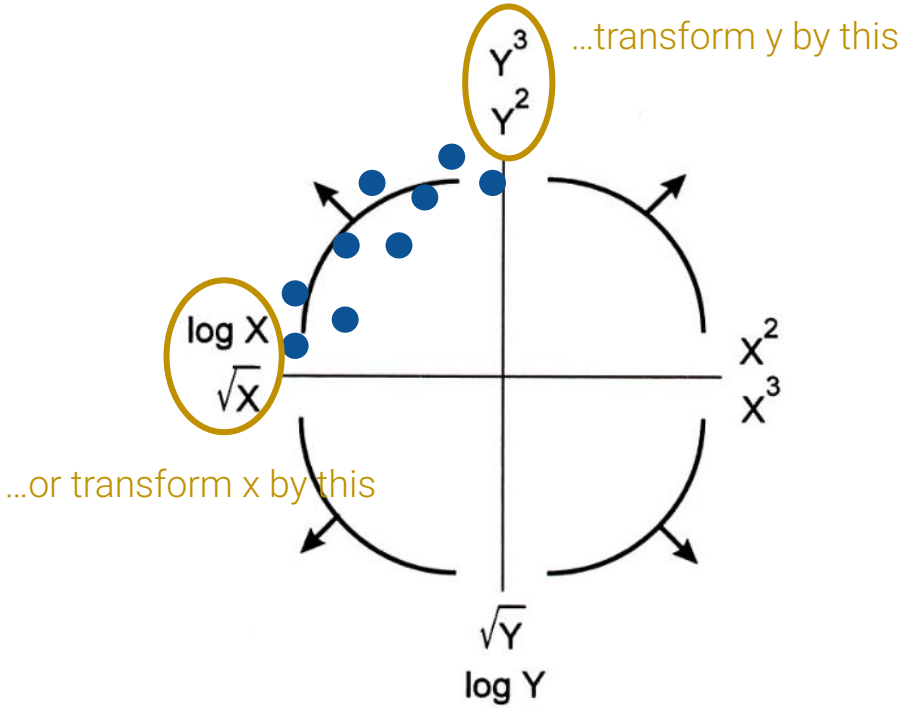
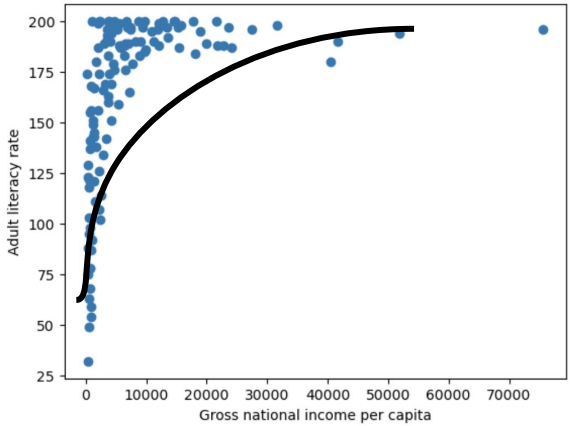
Tukey-Mosteller Bulge Diagram

If the data bulges like this…



Applying to the data from before:

Could have transformed y by y^2, y^3



Could have transformed x by log(x), sqrt(x)

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